An Intrinsic Big Bang Nucleosynthesis Fine-Tuning in Conserved-Charge Dark Matter from Relativistic MOND Clocks

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Abstract

A class of relativistic completions of Milgrom's dynamics keeps the modified-inertia sector *inside a clock* — a scalar φ entering only through $Q = n^\mu \partial_\mu \varphi$, with n_μ the unit gradient of a primordial clock field — and supplies cosmological dark matter as the conserved Noether charge of the clock-scalar's shift symmetry, $\varphi \to \varphi + \mathrm{const}$. This is an attractive architecture: the dark matter is then pressureless ($c_s^2 = 0$) and clusters like cold dark matter, it does not decay, and the same clock-internal structure lets the theory evade a preferred-frame post-Newtonian kill and a deep-MOND gradient instability. We prove that this architecture carries an **intrinsic Big Bang Nucleosynthesis (BBN) fine-tuning**. On a Friedmann–Robertson–Walker background the eliminated energy density is *quadratic* in the conserved charge C, $\rho = B + 3M^2H^2 - (M^2/3f^2)\,(A + C/a^3)^2$, so the same charge that sources the a^{-3} dust (cross term $\propto AC$) necessarily sources a stiff a^{-6} component (square term $\propto C^2$). The stiff term grows as a^{-2} relative to radiation toward early times; BBN bounds it to $\Omega_{\mathrm{stiff},0} \lesssim 4 \times 10^{-25}$. Demanding that the dust be the observed dark matter while the stiff term respects BBN forces $|C|/|A| \lesssim 3 \times 10^{-24}$ — a hierarchy of about twenty-four orders of magnitude, along a direction no symmetry protects. We prove the fine-tuning is **not removable** within the healthy class: the ghost-free degeneracy that keeps the clock's propagating sector free of a Boulware–Deser mode forces the constitutive potential $K(Q)$ to be exactly quadratic, which forces the density to be quadratic in C; a linear-in-charge (stiff-free) dust does not exist in the family; the tuning-free corner ($A = 0$) has *no* dark matter; and enlarging A merely relocates an equal $\sim$23-order tuning into the cosmological-constant sector. Every algebraic step is reproduced in exact symbolic arithmetic by committed scripts, and the load-bearing identities (the charge-quadratic density and the nonvanishing stiff coefficient) are additionally machine-checked in the Lean 4 proof assistant against its standard foundational axioms. **What is not claimed:** this is not a falsification — the theory is viable at the tuned point — and it is not a statement about whether nature is MONDian. It is a naturalness cost, one hierarchy worse than ΛCDM's, that any healthy conserved-charge MOND clock must pay unless a protecting mechanism absent from the present class is supplied. The construction whose fine-tuning is analysed here was developed collaboratively in the associated public repository; this paper isolates and proves the cost.

1 Setup: MOND inside a clock, dark matter as a Noether charge

We consider relativistic MOND completions of the form

$$S = \int \sqrt{-g}\Big[\tfrac{M^2}{2}R - \Lambda M^2 - K(Q) + F(Q)\,\Theta + M^2 a_0^2\, G(|V|/a_0)\Big] + S_m[g,\psi], \tag{1}$$

where a *clock* field T defines a unit timelike direction $n_\mu = -\partial_\mu T/\sqrt{-(\partial T)^2}$, $\Theta = \nabla_\mu n^\mu$ is its expansion, $Q = n^\mu \partial_\mu \varphi$ is the clock-projected derivative of the MOND scalar φ, $V_\mu = q_\mu{}^\nu \partial_\nu \varphi$ is the spatially-projected gradient, and M is the Planck scale ($G_\mathrm{N} = 1/16\pi M^2$). The scale a_0 is fixed by the de Sitter/Λ scale, $a_0 = c^2/2\pi L_\mathrm{dS}$, $L_\mathrm{dS} = \sqrt{3/\Lambda}$. The kernel $G(y) = y^2 + 2(1+y)e^{-y} - 2$ reproduces the exponential MOND interpolation, $G'(y)/(2y) = 1 - e^{-y}$. The MOND sector acts only through the clock and the spatial gradient; on a homogeneous background $|V| = 0$, so the G-term is cubic in perturbations and drops from the linear cosmology.

The action (1) is invariant under the shift $\varphi \to \varphi + \mathrm{const}$ (it depends on φ only through $\partial\varphi$). The corresponding Noether current is $J^\mu = (-K_Q + F_Q \Theta)\, n^\mu$, and its conserved charge density on FRW is

$$a^3\big(-K_Q + 3HF_Q\big) = C = \mathrm{const}. \tag{2}$$

This charge *is* the dark matter: as we recall below it redshifts as a^{-3} with zero pressure.

2 The theorem

Theorem 1 (intrinsic stiff partner) *Let (1) satisfy: **(a)** the propagating scalar sector is ghost-free on an expanding background, which by the background-independent velocity-Hessian degeneracy forces F affine, $F = fQ$, and $K_{QQ} = 3f^2/2M^2 = $ const (so K is exactly quadratic, $K = k_2 Q^2 + AQ + B$, $k_2 = 3f^2/4M^2$); **(b)** the cosmological dark matter is the conserved charge (2), $f \neq 0$. Then the Friedmann energy density is*

$$\rho = B + 3M^2H^2 - \frac{M^2}{3f^2}\Big(A + \frac{C}{a^3}\Big)^2 = \underbrace{\Big(B + 3M^2H^2 - \tfrac{M^2A^2}{3f^2}\Big)}_{\Lambda\text{-}const+back\text{-}reaction} \underbrace{-\frac{2M^2A}{3f^2}\,\frac{C}{a^3}}_{a^{-3}\ \ DUST} \underbrace{-\frac{M^2}{3f^2}\,\frac{C^2}{a^6}}_{a^{-6}\ \ STIFF}, \tag{3}$$

and the stiff coefficient $-M^2/3f^2$ is nonzero for any finite f. The dust piece is exactly pressureless ($w = 0$: the C/a^3 coefficient of the pressure vanishes identically); the stiff piece has $w = 1$ ($\rho_\mathrm{stiff} \propto a^{-6}$).

The elimination is elementary: (2) gives $Q = (3fH - A - C/a^3)/2k_2$ linear in C/a^3, and the lapse stress $\rho = K - QK_Q + 3HQF_Q$ is quadratic in Q because K is quadratic; hence ρ is quadratic in C/a^3. Because the coefficient of the square is the single number $1/(4k_2) = M^2/3f^2$, the dust and stiff pieces are the cross term and the square of one perfect square and cannot be separated. Every step is reproduced in exact `sympy` (script `L80_verify_fqtheta_dust.py`) and the two load-bearing identities — the decomposition (3) and $-M^2/3f^2 \neq 0$ — are machine-checked in Lean 4 (`Mondlean.lean`: `density_affine`, `stiff_coeff_ne_zero`), depending only on the standard axioms {`propext`,`Classical.choice`,`Quot.sound`}.

Theorem 2 (the fine-tuning) *Under Theorem 1, requiring the dust to supply the observed cold density $\Omega_c \simeq 0.264$ while the stiff term respects the BBN radiation-density bound forces*

$$\Omega_{\mathrm{stiff},0} \lesssim 4 \times 10^{-25}, \qquad \frac{\Omega_{\mathrm{stiff},0}}{\Omega_{\mathrm{dust},0}} = \frac{|C|}{2|A|}, \qquad \Longrightarrow \qquad \frac{|C|}{|A|} \lesssim 3 \times 10^{-24}. \tag{4}$$

The stiff density scales as a^{-6}, i.e. as a^{-2} relative to radiation, so the binding constraint is imposed at the earliest robust epoch, n/p freeze-out ($T \simeq 1$ MeV, $a \simeq 2.3 \times 10^{-10}$), where a $w = 1$ component must contribute below the effective extra-radiation bound $\Delta N_\mathrm{eff} \lesssim 0.5$. Propagating that bound to today gives $\Omega_{\text{stiff},0} \lesssim 4 \times 10^{-25}$ (script `L84_stiff_bbn_bound.py`, 12/12 checks; the number spans 10^{-25}–10^{-22} over $\Delta N_{\text{eff}} \in [0.3, 0.5]$, $T_{\text{BBN}} \in [0.07, 1]$ MeV). Because dust $\propto AC$ (linear) and stiff $\propto C^2$ (quadratic) share the single charge C, their ratio is $|C|/2|A|$, independent of M and f; matching the dust to Ω_c then requires the twenty-four-order hierarchy quoted. The bound is a_0-independent: a_0 enters only the galaxy-scale G-term, not the FRW stiff sector.

Corollary 1 (not removable) *No stiff-free dust exists in the healthy class, and the hierarchy cannot be relocated away. Specifically:* (1) *making K linear ($k_2 = 0$) removes the square but violates the degeneracy of Theorem 1 and leaves no propagating charge (no dust);* (2) *setting $f = 0$ removes the braiding but keeps a stiff term and, on an expanding branch, charge conservation forces $A = 0$ (no dust);* (3) *the tuning-free corner $A = 0$ has no a^{-3} dust at all;* (4) *enlarging $|A|$ (to reduce $|C|$ at fixed dust) is degeneracy-compatible but forces the constant B to cancel $M^2A^2/3f^2$ to precision $\delta_{CC} = 2(\Omega_\Lambda/\Omega_c)(|C|/|A|) \lesssim 1.6\times10^{-23}$, i.e. the same hierarchy moved into the cosmological-constant sector, and drives the decoupling sound speed $c^2_{\text{bare}} = 1 + A/2k_2Q$ to large magnitude. No shift or scaling symmetry of* (1) *caps C/A.*

Corollary 1 is established in exact arithmetic by scripts `L86_fine_tuning_assessment.py` (16/16) and `L87_stiff_free_dust.py` (19/19), which enumerate the linearisation attempts and the cosmological-constant relocation.

3 Discussion: what the cost is, and what it is not

The architecture analysed here is genuinely attractive, and the cost is worth stating precisely against its benefits. On the benefit side, the conserved-charge dust is pressureless and clusters like cold dark matter at linear sub-horizon order (its growth equation is ΛCDM's character-for-character, so $\sigma_8 = 0.811$ with no structure deficit), it cannot decay or annihilate (it is a Noether charge), and the clock-internal structure that generates it also lets the theory pass the preferred-frame post-Newtonian gate that excludes AeST and evade a deep-MOND gradient instability. The present result shows the same conserved charge that carries all of this also carries a stiff early-time partner, and that BBN prices it at $|C|/|A| \lesssim 3 \times 10^{-24}$.

This is a naturalness statement, not a falsification. ΛCDM fixes Ω_Λ and Ω_c as two relic/initial-condition numbers, neither tuned against a pathology; the present class fixes the same two *plus* the extra hierarchy, along an unstable direction. Framing C as a conserved integration constant (initial data) softens but does not remove the cost, since conservation forbids dynamical relaxation and matching Ω_c re-imports a coefficient tuning. A protecting mechanism — a symmetry under which C^2 is forbidden while AC survives, or a construction in which the dust is not the clock's kinetic charge — would evade the theorem; none exists in the class (1), and the only stiff-free alternative found (an externally added cold-dark-matter fluid) abandons the one-action structure and re-introduces the transmission overshoot such completions were built to avoid.

4 Reproducibility

Every numerical and algebraic claim is produced by a committed, runnable script with checks that can fail, in the public repository under `fable_independent_2026/`: `L80_verify_fqtheta_dust.py` (the density decomposition), `L84_stiff_bbn_bound.py` (the BBN bound), `L86_fine_tuning_assessment.py` (non-removability, cosmological-constant relocation), `L87_stiff_free_dust.py` (no linear-charge dust in the healthy class). The two load-bearing identities are machine-checked in Lean 4 under `fable_independent_2026/lean_2026/Mondlean.lean`. All dimensional numbers are carried on both a_0 footings ($9.3619 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ and $1.1279 \times 10^{-10}\,\mathrm{m\,s^{-2}}$); the fine-tuning is a_0-independent. AI-assisted research draft; not peer reviewed.

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